

Data science and GIS-based system analysis of transit passenger complaints to improve operations and planning

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ABSTRACT

Transit user complaints support system resilience by serving as a data source for service improvements. This study shows how geographic information system (GIS)-based analysis, econometric models, and latent class analysis can improve system-wide understanding of passenger complaints. The analyzed dataset consists of 718 passenger complaints concerning the operation of municipal lines in Jerusalem as the study region. The analytical methods consists of GIS-based analysis and statistical modeling: mapping, recursive bivariate probit estimation, negative binomial model estimation, and latent class analysis. The GIS-based analysis showed that the spatial distribution of complaints changes over time as a function of service disruption type and geographical area. The recursive bivariate probit model results indicated that the most acute sources of frustration are service problem recurrence and monetary loss, with the former caused by overcrowding, delays and line cancellations. The negative binomial model results shows that the number of complaints increases with an increase in the passenger boarding to bus arrivals ratio. Latent class analysis reveals that, in terms of both prevalence and customer frustration, overcrowding delays and line cancellations are the most acute problems in the study region. The proposed interface between transit complaints and GIS databases can readily be implemented by transport operators and authorities.

1. Introduction

Smart cities and transport systems rely on data from individual customers to understand citizens' needs and generate solutions to urban problems and traffic bottlenecks (Lim et al., 2018). Civic complaints and other user-oriented big-data sources are essential for preventive local administration in smart cities; hence, recent studies have aimed to determine how to use civic complaint data to improve city management. Example cities are New-York (Bansal and Toshniwal, 2016), Busan (Lim et al., 2018), Jinju-Si (Won and Yoo, 2016), and Gyeongsangbuk-do (Jun and Chung, 2016). Smart transport systems are an integral part of smart cities (Lim et al., 2018). The development of smart passenger service technologies has opened up new opportunities to improve the two-way communication between service providers and passengers and address passenger needs (Hensher, 2017). Since transit is a complex, open system susceptible to service disruptions, passenger complaints could be integrated into smart communication technologies to improve transit

planning.

Studies showed that passengers often experience service shortages and failures and are willing to be proactive in communicating service issues (Mikhaylov et al., 2016; Sarker et al., 2019). In a survey completed by bus passengers in Kaliningrad, Russia (Mikhaylov et al., 2016), 42% of the respondents experienced at least one negative incident, and 14–28% were willing to file a formal complaint, depending on the communication method (e.g., phone, e-mail, or social media). In a recent study conducted among bus passengers in Innsbruck, Austria, where the perceived service quality was high (Sarker et al., 2019), 25% of the respondents experienced line cancellations or non-functioning ticket machines, 16% experienced bus technical failures, and 36% experienced missed connections. 16–20% of the respondents were willing to complain to the operators about critical incidents (Sarker, 2019); hence, transit passengers are likely to become proactive and involved system users, who can provide valuable voluntary data.

The usefulness of efficient system-wide handling of transit

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complaints is threefold. First, complaints have direct implications for transit ridership and passenger loyalty. Friman et al. (2001) showed that transit satisfaction was linked to the number of adverse critical incidents that passengers recalled. Joewono and Kubota (2007) showed that unpleasant Paratransit service incidents negatively influenced passenger satisfaction and loyalty. Friman (2004) and Pedersen et al. (2011) showed that communicating hypothetical positive and negative transit experiences could trigger a higher or lower perceived level of satisfaction, respectively. Johnson (2001), who studied customer service departments in organizations, showed that good complaint culture led to process improvements and customer satisfaction; hence, transit providers should strive to analyze complaints to improve passenger satisfaction and transit use efficiency.

Second, according to the Hirschman exit, voice, and loyalty theory (1970), filing formal complaints is a proactive response that aims to communicate problems and induce change. This theory was supported by the findings of Chou and Kim (2009), whose statistical model results showed that, among high-speed rail passengers in Taiwan, higher satisfaction was associated with a greater tendency to file formal complaints, which translated into higher consumer loyalty. Chou and Kim (2009) and Lai and Chen (2011) viewed complaint handling as contributing to overall service levels and key to maintaining consumer loyalty.

Third, complaints can aid transit planning and management, by serving as performance indicators and exposing service problems (Edvardsson, 1998; Friman et al., 1998). Bus stop skipping, delays, and overcrowding that surface through complaints can be examined and verified using big data from bus-mounted GPS and automated passenger counters (Yap and Munizaga, 2018). Complaints can help transport authorities and non-profit transport user organizations (e.g., London TravelWatch) to monitor service levels and identify specific issues that remain unresolved (Thao et al., 2017). Complaints are necessary for identifying and rectifying service issues for under-represented user groups, such as mobility-impaired people (Major and Hubbard, 2019). Geocoding transit complaints can help to identify the spatial distribution of complaints (Liu and Yen, 2016; Won and Yoo, 2016).

To date, only a handful of studies have formally analyzed formal transit complaints. Friman et al. (1998) and Friman and Edvardsson (2003) used manual content analysis to review 200 complaints filed with a transit company in Gothenburg, Sweden, as indicators of adverse critical incidents. They found that the most problematic issues were impolite employee behavior, lack of punctuality, lack of information, technical malfunctions, poor infrastructure design, and crowded vehicles. By comparing the complaints to an on-board survey of critical incidents, Friman and team (1998) showed that complaints represent a broad range of consumer transit experiences. The complaint themes were similar to concerns raised in passenger surveys in other regions (Joewono and Kubota, 2007; Mikhaylov et al., 2016). Orthaber and Márquez-Reiter (2011) analyzed complaint calls to a Slovenian transit service provider, finding that complaints followed an intricate linguistic pattern and were used to air negative emotions, making it difficult for the company to handle the complaints. Wittman (2014) showed that airline complaint rates correlated with airline performance in terms of delayed flights, mishandled baggage, and overbooking.

Passenger complaints may involve lengthy and fuzzy texts, received by phone or e-mail, which are manually analyzed by transit operators worldwide (Orthaber and Márquez-Reiter, 2011). Each complaint is handled separately, in a lengthy and costly procedure that is susceptible to mistakes and unable to provide system-wide solutions (Liu and Yen, 2016). These methods impede bulk complaint handling and system-wide service improvements. Better solutions are necessary to improve the use of passenger complaints as a data source. Congosto and coworkers (2015) showed that Twitter messages regarding transit service events could be translated into a heat map of location-based service performance. Similarly, Liu and Yen (2016) analyzed the existing bus passenger complaint management process in Taiwan and suggested

methods of improving complaint-handling time and quality, including the generation of heat maps to identify complaint hotspots.

The present study proposes a novel approach to analyzing passenger complaints by using combined data science methods, including geographical information system (GIS)-based analysis and statistical modeling. The study expands the research of Liu and Yen (2016) by conducting a system-wide analysis of complaints to improve complaint-handling processes and manage service problems from a system perspective.

The dataset for the analysis consists of the entire complaint dataset (718 complaints) relating to service operation (accessibility, reliability, and tangibles) received by the Ministry of Transport (MOT) for West-Jerusalem, Israel, during the period 2017–2019 concerning the municipal bus service. We conducted three types of statistical analysis. First, we use geographical information systems (GIS), to enable a system-wide analysis of complaints and improve problem-solving. The analysis shows the spatiotemporal distribution of service problems by disruption type and bus line. Second, a recursive bivariate probit estimation reveals the antecedents of passenger frustration as a function of recurrent service issues and complaint characteristics. Third, latent cluster analysis uncovers bulk complaints and helped to prioritize them according to their prevalence and importance.

The article structure is as follows. Section 2 presents the case study in terms of complaint-handling processes and the transit system in the study region. Section 3 provides details of the analyzed dataset. Section 4 describes the data analysis methods. Section 5 presents the results. Section 6 presents the discussion and conclusions.

2. Case study

2.1. Complaint-handling processes in the study region

The MOT mainly conducted passenger complaint handling via regular mail, phone, and fax services during the 1990s. The number of complaints was limited and complaints were handled manually. Following technology advances and transit privatization, the MOT established an internet-based system for complaints, information requests, and suggestions. The system initially included an open textbox, which evolved to its current four-stage submission format in 2018. The first page requests identity and contact information. The second page addresses the relevant travel mode (bus, taxicab, train, or other) and the issue (e.g., line cancellation, overcrowding, skipping a stop). The third page establishes the service provider, line number, driver's name, date and time of the event, bus line direction, boarding stop, and a detailed textual description of the event using an open textbox. The fourth page includes a space for uploading text files and photographs.

At present, there are three parallel submission channels for formal complaints: directly to transport providers, directly to the MOT, and to a non-profit passenger organization named 15 min. The channels are independent, and the complaints are handled separately. Following technological improvements and increased public awareness, the number of nationwide complaints to the MOT nearly doubled, from 25,726 yearly complaints in 2016 to 37,000 in 2018 and 47,130 in 2019. The current number of nationwide monthly complaints ranges between 2000 and 5,000, and there were 3931 average monthly complaints during 2019. Of the complaints, 52% concerned schedule deviations, including line cancellations, delays, and early arrivals. About 16% of the complaints included requests for service changes, such as additional stops, route changes, and bus frequency increases. Only 3.5% of the complaints concerned driver behavior. 47.4% of the complaints were sent to the control department and roughly 44.5% of those (21.1% of the submitted complaints) led to fines imposed by the MOT on the transit service providers (Ministry of Transport, 2019).

The complaints are handled by a team of 18 designated MOT employees, who investigate the complaints and provide customized answers. The complaint handling consists of classification and assignment,

identifying unusual or urgent issues, investigation of the complaint by the relevant parties, and responding to the passengers. The complaints are handled on a first-come-first-served basis, with priority given to complaints about line cancellations and delays. The maximum response time for complaints regarding line cancellations and delays is about 5 days and, for other issues, 14–20 days. The complaints are handled individually, with no system-wide overview. During 2019, on average, 3500 complaints were handled by the MOT. The average complaint processing time was 8.45 days, and, hence, the main complaint-handling expenses related to human resources (e.g., salaries).

2.2. Analyzed municipal bus service

Jerusalem is the largest city in Israel, with 882,700 inhabitants. Detailed in-depth overviews of the transit planning and demand in Jerusalem city were provided by Feitelson and Cohen-Blankshtain (2018) and Kerzhner et al. (2018). The municipal bus service in West Jerusalem, the study area, is operated by one of the major Israeli transit providers, which operates 33% of the transit in the five metropolitan areas of Israel: 3500 buses along 900 routes and 265 million yearly nationwide passengers. Within the study area, the provider operates 73 bus lines, characterized by four service types: express lines along main radial routes with bus priority lanes, radial lines connecting main destination points in the city along main routes, city lines connecting primary and secondary destinations, and feeder lines to the light rail transit (LRT) stations. The bus fleet operating the municipal lines comprises approximately 700 buses with roughly 7240 daily bus arrivals on regular weekdays. There are approximately 600,000 daily boarding passengers in the service area, with 160,000 connecting with the LRT system and 450,000 boarding municipal buses. During 2018, the Jerusalem municipal bus system served 57.4 million yearly passengers (Ministry of Transport, 2018).

3. Data

The analyzed dataset includes complaints relating to planning and service operation issues (reliability and tangibles), which are managed by the MOT. We did not analyze complaints regarding driver behavior, due to their scarcity—only 2.6% of the complaints submitted to the MOT in the study area concerned driver behavior. The analyzed dataset includes 718 complaints submitted between January 2017 and November 2019 regarding 70 city lines operating in West Jerusalem. The complaints were communicated by phone, e-mail, or through an online complaint form and were recorded in the MOT's customer relations management (CRM) system. The CRM system held 853 complaints relating to the designated service area. After removing complaints referring to school lines and complaints without textual messages, 734 complaints remained, of which 718 comprised valid data for coding.

The complaint data in the CRM system included the time and date of the complaint, the line number and direction, and a textual description of the complaint. The data was provided for the analysis in anonymized form due to privacy considerations. The 1981 Privacy Act does not allow personal information relating to public complaints to be revealed. All data management actions concerning public complaints are subject to the Privacy Act and can only be analyzed in an anonymized form. In addition, the MOT regulations allow contact with passengers submitting complaints only for complaint-handling purposes. Notably, the 1981 Privacy Act does not allow data analysis related to individual socio-economic factors, which was a limitation of the analysis. Fig. 1 shows the distribution of the number of words per complaint. There were an average of 121.17 words per complaint, with a standard deviation of 70.17 words. Of the complaints, 45% were less than 100 words and 85% were under 200 words.

The complaints contained a great deal of unnecessary, irrelevant, unclear, or imprecise information; lacked event location and time identifiers; or complained about multiple lines and various service issues

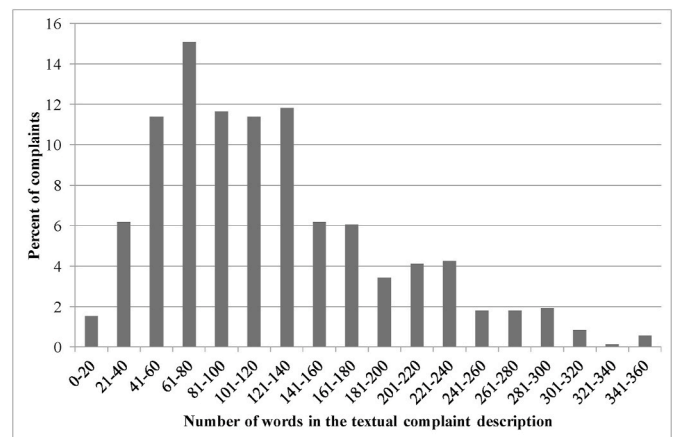


Fig. 1. An example of a complaint word distribution.

within the same textual description. The analyzed text was rich and varied, expressing multiple complaints, mixed complaints, sarcastic comments, and irony that were difficult to encompass with simple classification codes.

An example of a complaint combining three different issues is shown in Fig. 2: low service frequency, high rate of line cancellations, and overcrowding. The description also complains about social injustice in transit provision and frustration is explicitly expressed in the lengthy text. Indirectly, it expresses satisfaction about the high service frequency and low overcrowding level of other lines. The person who filed the complaint explicitly requests an answer. Content analysis of key words revealed redundant information in the complaint and problems discerning the relevant bus lines, the specific issues encountered, and the location of the problem. The description highlights the need for efficient recoding to use the complaint data with the GIS database.

“Hello, I would like to express my **frustration** with a certain phenomenon. I am a student and a resident of Jerusalem for the past six years, taking bus **lines** 19, 27, and 42 to the university campus. I always understood, according to your explanation, that there are not enough **drivers**; therefore the bus **frequency** is every 15 min (the buses **arrive** every 30 min, and many buses **do not arrive**). Until yesterday (08.10.17), that was the situation according to my understanding. Yesterday, I rode the **bus line** 402 from Bnei-Brak to Jerusalem. To my surprise, the bus **arrived** every 3 min (according to the internet website, it should arrive every 5 min), so it was excellent (for some passengers). When I arrived at the central bus terminal in Jerusalem, I **waited** for **lines** 25 or 67 to Jerusalem. While I was **waiting**, I encountered the following interesting situation. While I **waited over 20 min**, **line** 3 to the old city center arrived almost every minute, with only one passenger boarding each time. Now to my complaint. How can it be that for **lines** 19, 27, and 42, there are no **drivers** and buses therefore **arrive** every 15 min (every 30 min actually), and students are **cramped** and almost unable to breathe, but **lines** 402 and **lines** 3 have a **frequency** of 3 min? Do I understand from this situation that some populations are preferred over others? I only understood it yesterday, but I wonder, for how long has the situation been going on, and why are some populations preferred over others? I will be happy to receive a response.”

To perform the GIS-based and statistical analysis, it was necessary to recode the textual items into numeric variables. The initial classification of the CRM system was based on the complaint title and did not fully encompass the complaint; hence, the full text of each complaint was manually coded. The complaints were uploaded into a database, including the line number identifier, eight binary variables representing service disruptions, and five binary variables relating to the user experience. Service disruptions consisted of line cancellations, delays, overcrowding, cleanliness issues, inefficient planning, long walking distances, and requests for new services. The user experience variables

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Fig. 2. An example of a textual complaint description. Keywords are printed in bold letters.

referred to expressed frustration, tangible losses (monetary or time), experiencing recurrent service problems, general service dissatisfaction, and positive attitudes indicating trust. Because multiple complaints for the same lines were common, but complaints addressing different lines were rare, complaints addressing multiple lines were treated as separate records. Due to the textual complexity, the variables were manually extracted from the text, but for larger samples, computerized algorithms should be developed.

Table 1 presents the frequency in the dataset of service disruptions by type and user experience. 9.2% of the complaints detailed one type of service disruption, 48.0% detailed two types of service disruptions, and 42.8% detailed at least three types of service disruptions. The most frequently reported incidents were about low bus frequency and overcrowding. Users were three times more likely to express frustration than complain about monetary or time loss. Interestingly, despite the low rate of reported user satisfaction (5.6%), and the high rates of expressed frustration (42.7%), recurrent issues (51.1%), and non-responses (34.7%), 82.9% of the complaints expressed confidence that the reported issues could be resolved. These results confirm that formal complaints are proactive responses, communicating problems in order to induce change.

4. Analytical methods

4.1. GIS mapping

Fig. 2 describes the data sources and data processing. The mapping procedure involved integrating four different databases into a single GIS database. The four databases were the Jerusalem Master Plan Transport Model, the MOT transit database, smartcard usage data, and the MOT's

CRM system.

The data from the Jerusalem travel behavior simulation model included the motorization level and modal split at the traffic analysis zone (TAZ) level, and the level of service according to the volume to capacity (V:C) ratio as detailed in the American Highway Capacity Manual (2000). The MOT transit database included line routes, bus stops, and schedules in GTFS format, which allowed the number of daily bus arrivals per bus stop to be calculated. Smartcard usage data provided daily boarding numbers per bus stop. In Israel, smartcard validation is conducted upon boarding and, therefore, the number of alighting passengers and bus overcrowding could not be calculated. The MOT's CRM system assigns a passenger complaint to a unique line identifier that matches the GTFS database. The textual complaints were coded alpha-numerically, in a GIS-compatible format, and the analysis was conducted on the MOT GIS-based TransCad platform. The MOT and the Jerusalem Master plan used different transport modeling software (TransCAD and EMME/2), with different spatial resolutions and geographical representations of the road network (geometric coding in EMME2 and GIS-based coding in TransCAD); thus, we conducted map matching to integrate the sources from the two platforms. The GIS-based road network on the MOT's platform was used to calculate network-based distances, due to its geographic accuracy.

4.2. Bivariate probit estimation

Frustration and problem recurrence were modeled as a function of the complaint characteristics using a recursive bivariate probit model, which represented seemingly unrelated and recursive nonendogenous processes (Heckman, 1978). In the recursive case, in addition to being influenced by similar explanatory variables, the outcome of one process served as an endogenous variable for the other process. In the recursive case, the directionality of the causality should be a-priori assumed, although unless panel data or a treatment and effect are employed, reverse causality cannot be ruled out Fig. 3.

The structural model consists of two latent variable equations. The recursive model form is as follows (Winkelmann, 2012):

$$\begin{aligned} y_1^* &= x\beta + \alpha y_2 + \varepsilon_1 \\ y_2^* &= z\gamma + \varepsilon_2 \end{aligned}$$

The vectors of observed explanatory variables z and x may share similarities in the observed variables, and the error terms ε_1 and ε_2 are not necessarily independent. The observed binary outcomes are:

$$y_{k=1,2} = \begin{cases} 1 & y_k^* > 0 \\ 0 & \text{otherwise} \end{cases}$$

Table 1
Frequency of service disruptions by type and user experience.

Service problems		User experience	
Variable	Percentage	Variable	Percentage
Line cancellations	13.0	Expressed frustration	42.7
Delays	10.0	Tangible loss	14.2
Overcrowding	49.0	Experiencing recurrent problems	51.1
Low frequency	58.2	High level of service perception	5.6
Poor cleanliness	2.2	Positive attitude indicating trust	82.9
Planning inefficiency	9.9		
Long bus stop walking distances	9.7		
Requests for new services	6.0		

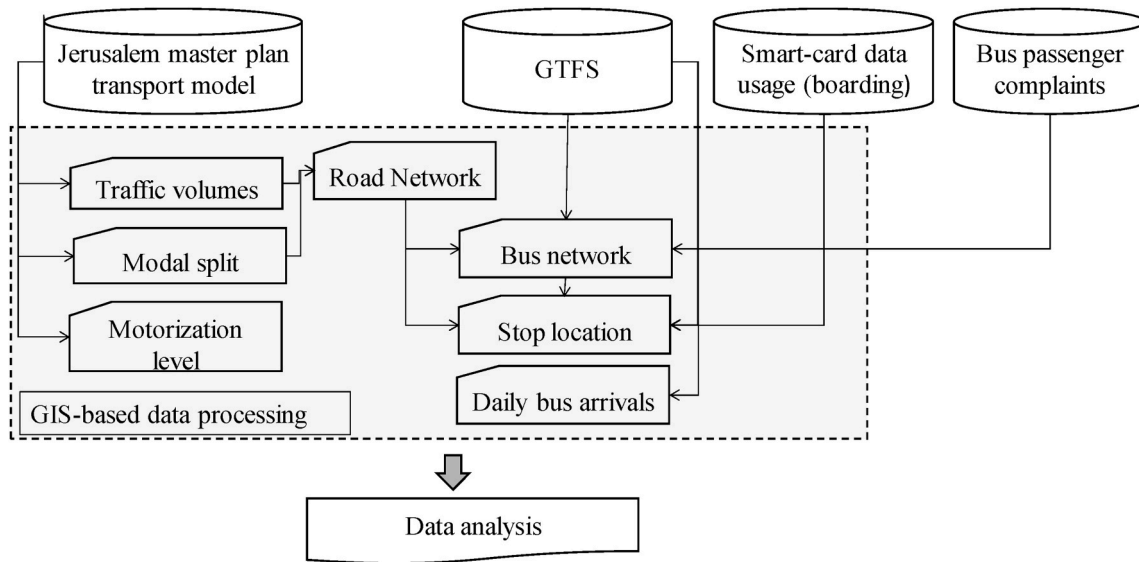


Fig. 3. GIS-based data processing for the analysis.

In the case of full observability, the joint distribution of the two variables includes all the four outcome combinations. Here, the distribution follows a Gaussian distribution. The probability is written as:

$$\begin{aligned}
 P(y_1 = 0, y_2 = 0) &= C[\Phi(-x\beta), \Phi(-z\gamma)] \\
 P(y_1 = 1, y_2 = 0) &= C[1, \Phi(-x\beta)] - C[\Phi(-x\beta), \Phi(-z\gamma)] \\
 P(y_1 = 0, y_2 = 1) &= C[\Phi(-x\beta - \alpha), 1] - C[\Phi(-x\beta), \Phi(-z\gamma)] \\
 P(y_1 = 1, y_2 = 1) &= 1 - C[\Phi(-x\beta - \alpha), 1] - C[1, \Phi(-z\gamma)] \\
 &\quad + C[\Phi(-x\beta - \alpha), \Phi(-z\gamma)]
 \end{aligned}$$

Assuming n observations, the likelihood function is as follows:

$$\prod_{n=1}^N P(y_{n1} = 1, y_{n2} = 1)^{y_{n1}y_{n2}} \times P(y_{n1} = 1, y_{n2} = 0)^{y_{n1}(1-y_{n2})} \times P(y_{n1} = 0, y_{n2} = 1)^{(1-y_{n1})y_{n2}} \times P(y_{n1} = 0, y_{n2} = 0)^{(1-y_{n1})(1-y_{n2})}$$

The joint probabilities depend on the selected copula and the parameter vectors $(\alpha, \beta, \gamma, \text{ and } \theta)$, where θ is the dependence parameter of the copula function. The interpretation of the recursive bivariate probit correlation parameter does not capture the correlation between the analyzed variables. Once the effect of the endogenous variable is taken into account, the correlation between the error terms is not necessarily of the same sign as the endogenous relationship (Fillipini et al., 2018). If the copula is a fully parametric function, a consistent and asymptotically normally distributed estimator of the parameters can be obtained through maximum likelihood estimation. The R package GJRM served for the model estimation (for documentation see Marra and Radice, 2017).

4.3. Negative binomial model estimation

The number of complaints per bus stop is a count variable analyzed using a negative binomial model. In the context of the number of transit complaints per bus stop, the assumptions of the Poisson model, (the independence between event occurrences and the equality of the conditional mean and variance) are too restrictive. Passenger complaints, particularly concerning unreliability (e.g., delays and line cancellations), could be dynamically interdependent events affected by similar spatial characteristics at the location of the complaint. In addition, overdispersion is expected due to bus stop location heterogeneity.

The negative binomial model assumes that the event rate λ_i follows a gamma distribution with mean φ_i and precision parameter v_i (Cameron and Trivedi, 1986):

$$f(\lambda_i) = \frac{1}{\Gamma(v_i)} \left(\frac{v_i \lambda_i}{\varphi_i} \right)^{v_i} \exp \left(-\frac{v_i \lambda_i}{\varphi_i} \right) \frac{1}{\lambda_i}$$

The probability density is as follows:

$$P(y_i) = \frac{\Gamma(y_i + v_i)}{\Gamma(y_i + 1)\Gamma(v_i)} \left(\frac{v_i}{v_i + \varphi_i} \right)^{v_i} \left(\frac{\varphi_i}{v_i + \varphi_i} \right)^{y_i}$$

$$E[y_i] = \varphi_i$$

$$\text{var}[y_i] = \varphi_i + \frac{1}{v_i} \varphi_i^2$$

We consider the following relationship between the distribution parameters and the explanatory variables:

$$\varphi_i = e^{x_i \beta}$$

$$v_i = \frac{1}{\alpha} (e^{x_i \beta})^k$$

For a positive parameter $\alpha > 0$ and a constant k that sets the relationship between the variance and the mean of the distribution.

$$\text{var}(y_i) = E[y_i] + \alpha (E[y_i])^{2-k}$$

Setting the constant k to unity ($k = 1$) indicates a constant variance: mean ratio (negative binomial type I), whereas equating k to zero ($k = 0$) indicates that the variance:mean ratio is linear for the mean (negative binomial type II). In the current study, we estimated a negative binomial type II model.

The negative binomial model was estimated by using the Python Statmodels module for the general linear regression. The module is validated and verified against existing statistical packages (Seabold and Perktold, 2010).

4.4. Latent class analysis

As specified in the data section, most (90.8%) of the complaints specified two or more service issues. Simple one-to-one classification could not accommodate the complexity of the textual complaint; instead, latent class analysis (LCA) was chosen for complaint pattern recognition to investigate whether we could identify and prioritize complaint bulks. The LCA was performed using the categorical indicators corresponding to the complaint characteristics as explanatory variables. Complaint recurrence and frustration were the target variables. Latent class analysis is a model-based clustering or finite-mixture

clustering method. We selected LCA over alternative clustering methodologies due to the existence of an underlying statistical model, LCA's ability to represent similarity across clusters, and goodness-of-fit criteria that determined the optimal number of clusters. Nevertheless, other clustering methods could have been used, such as artificial neural networks (ANNs) to create self-organizing maps (Kohonen maps) that can be nicely visualized, classification trees (gradient boosting decision trees and random forests), and swarm-intelligence for semi-supervised classification.

The latent class model, described in detail by Lanza and colleagues [29], is summarized as follows. K latent subgroups can be inferred from

$j = 1, \dots, J$ observed variables, each of which has $r_j = 1, \dots, R_j$ response categories. C represents the latent variable with latent classes $c = 1, \dots, K$, and x represents the vector of responses to the J variables $x = (r_1, \dots, r_J)$. $I(x_j = r_j)$ is an indicator function that equals 1 when the response to variable $j = r_j$, and equals 0 otherwise. The probability of observing a particular response pattern is as follows [29]:

$$P\{X = x\} = \sum_{c=1}^K \gamma_c \prod_{j=1}^J \prod_{r_j=1}^{R_j} \rho_{jr_j|c}^{I(x_j=r_j)} \quad (1)$$

Latent class analysis can be employed with or without considering

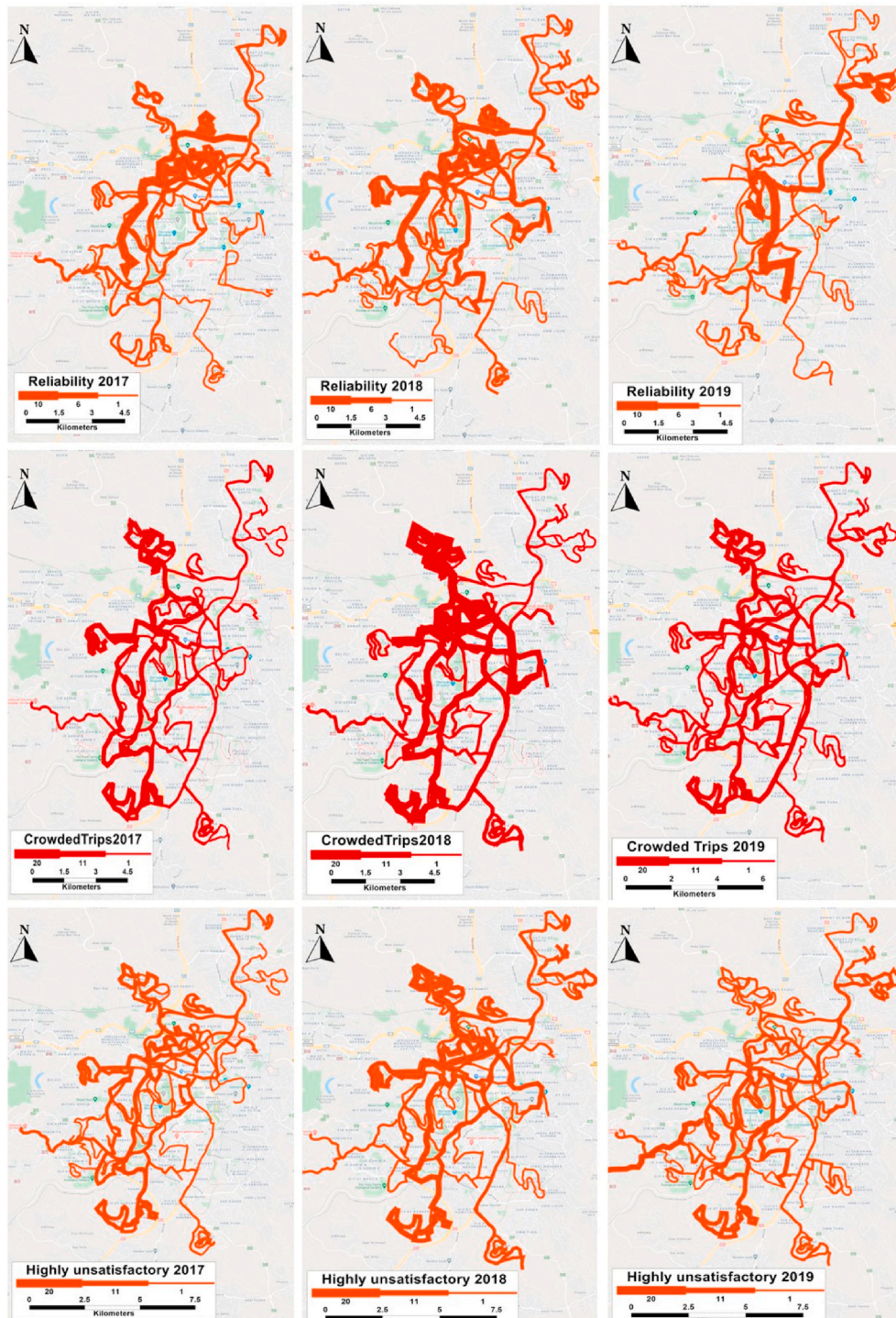


Fig. 4. Service unreliability and frustration by year (2017–2019). Top panels: unreliability. Middle panels: crowded trips. Bottom panels: frustration levels.

distal outcomes. Frustration and problem recurrence were treated as distal outcomes, because they resulted from the complaint characteristics.

In LCA with a distal outcome, the density assumes conditional independence between the distal outcome Z and the latent class membership C ; hence, the density function reads $f\{Z = z|C = c\}$. We can determine the desired distribution of $Z|C$ by applying Bayes' theorem [29]:

$$f\{Z = z|C = c\} = \frac{f\{Z = z\}f\{C = c|Z = z\}}{f\{C = c\}} \quad (2)$$

The density function $f\{C = c\}$ is determined by the LCA model and $f\{C = c|Z = z\}$ is determined by the LCA model with the distal outcome Z included as a covariate. The density function $f\{Z = z\}$ depends on the distribution of Z . When Z is a binary variable it can be incorporated as a

covariate in the LCA model. Then, the density function $f\{Z = z|C = c\}$, can be expressed as [29]:

$$P\{Z = z|C = c\} = \frac{P\{Z = z\}e^{\beta_{0c} + \beta_{1c}z}}{P\{C = c\}(1 + \sum_{c'=1}^{K-1} e^{\beta_{0c'} + \beta_{1c'}z})} \quad (3)$$

Using this approach, $P\{Z = z\}$ is estimated from the empirical distribution of Z , namely from the proportions in the observed data. The estimates for $\{\beta_{0c}, \beta_{1c}; c = 1, 2, \dots, K-1\}$ are provided by the LCA with covariates model. The marginal distribution $P\{C = c\}$ can be obtained by multiplying $P\{C = c|Z = z\}$ by the marginal distribution $P\{Z = z\}$. Thus, $\Pr\{Z = z|C = c\}$ can be estimated given the estimates for $P\{Z = z\}$, $P\{C = c\}$, and $\{\beta_{0c}, \beta_{1c}; c = 1, 2, \dots, K-1\}$.

We estimated the LCA model using the robust maximum likelihood estimator (expectation–maximization algorithm). The number of latent classes was determined on the basis of the Bayesian information

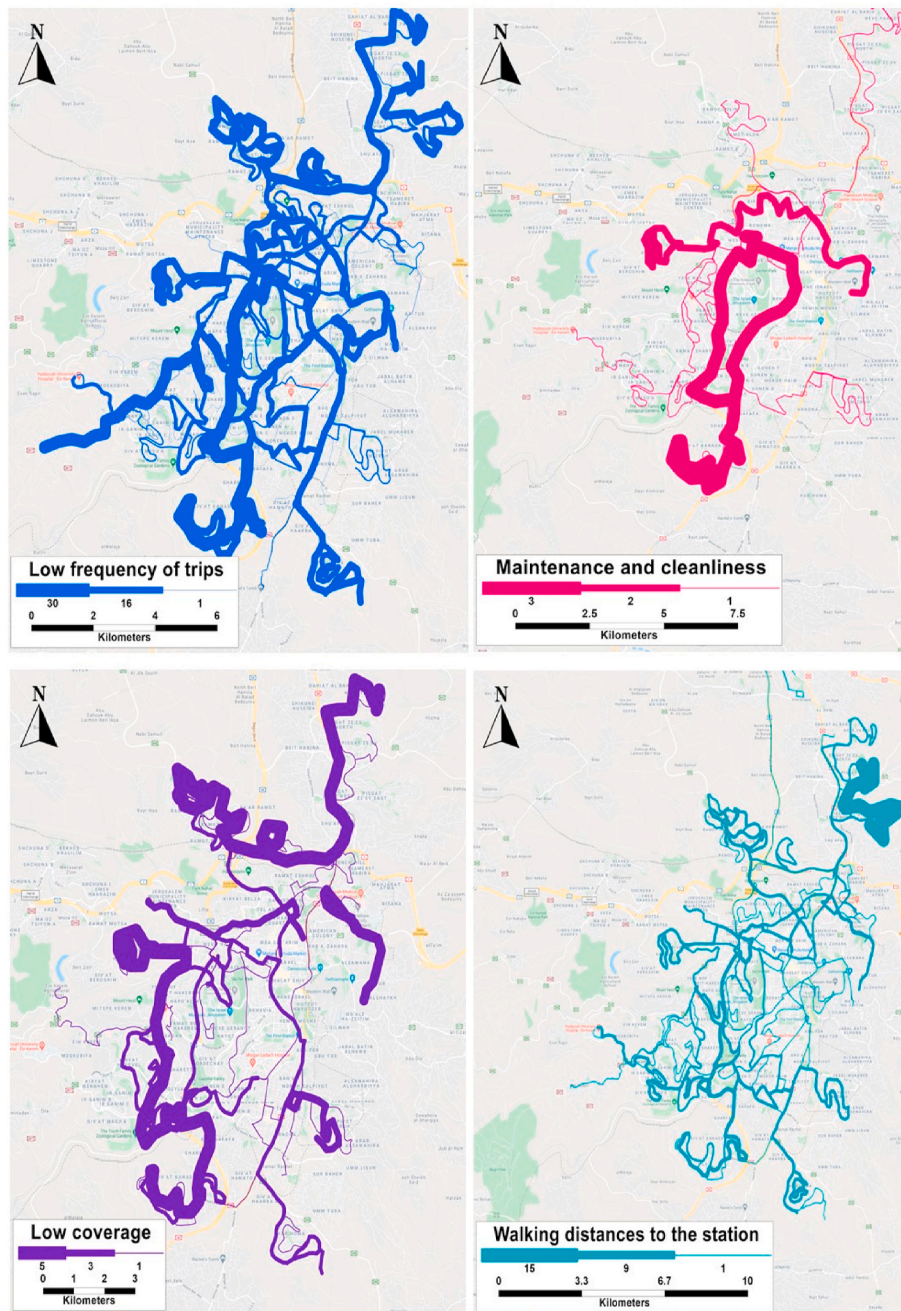


Fig. 5. The geographical spread of complaints by service disruption type.

criterion (BIC).

5. Results and discussion

5.1. Frustration and problem recurrence motivators

5.1.1. Mapping

Fig. 4 displays the changes in reliability and levels of frustration over the years 2017–2019. The three top panels represent the number of complaints about service unreliability (line cancellations and delays), the three middle panels show the number of complaints about overcrowding, and the three bottom panels show the number of complaints that expressed frustration. The figure shows that unreliability complaints had a wider spatial distribution in 2018 than 2017, but the spatial distribution in 2019 indicates that the problems encountered in 2017–2018 were mostly resolved. According to a consultation with the MOT, a severe driver shortage during 2018 led to line cancellations and delays. Following those events, system-wide changes were made, including (1) increasing the efficiency of line operations; (2) a 5% system-wide frequency reduction to cope with the driver shortage and reduce delays; (3) quality assurance and greater operational control by both the MOT and the transit provider; and (4) efforts to recruit more drivers, supported by monetary incentives. These changes considerably reduced the number of unreliability complaints, but significantly increased the number of bus overcrowding complaints. During 2019, the numbers of complaints about both crowded buses and service unreliability decreased, due to driver recruitment and gradual bus frequency increase that helped to maintain service reliability. Frustration remained widespread over the three years in agreement with the bus overcrowding complaint pattern. The map shows that, in 2019, frustration was somewhat relieved, mainly in northwestern neighborhoods, while it increased in southwestern neighborhoods.

Fig. 5 shows the spatial distribution of encountered service issues (low service frequency, cleanliness issues, requests for additional transit services, and long walking distances to bus stops) across the service area during the study period (2017–2019). The figure shows that low frequency was a widespread problem. By contrast, cleanliness, walking distances, and new service requests were local problems limited to specific lines and traffic analysis zones; they were thus less acute and could more easily be resolved by pinpoint solutions.

5.1.2. Frustration and problem recurrence—recursive bivariate probit model

In 35.4% of the complaints, both frustration and problem recurrence were mentioned, whereas in 40.0% of the complaints, neither issue was expressed. In 24.7% of the complaints, either frustration or problem recurrence was mentioned. A Lagrange multiplier test detected endogeneity; thus the two equations needed to be simultaneously estimated. First, the same problem characteristics can drive both frustration and problem recurrence. Second, both causality and reverse causality are possible. The intuitive assumption was that problem recurrence induced higher levels of frustration; however, we could not rule out the possibility that higher frustration levels led to mentioning problem recurrence. Therefore, three model specifications were tested: bivariate probit with seemingly unrelated equations, recursive bivariate probit assuming direct causality, and recursive bivariate probit assuming reverse causality. Tests for non-nested models showed that the recursive model, with problem recurrence explaining frustration (direct causality), was far superior to the other two specifications at the 0.001 significance level. Table 2 presents the best model specification.

The model results show that compared to other service issues, bus cancellations and delays increase the probability of complaints about problem recurrence. It also shows that general satisfaction with the service level, reduces the probability of complaints about problem recurrence. Problem recurrence, time or monetary losses, and planning inefficiency are associated with an increased probability of frustration.

Table 2

Estimated recursive bivariate probit model parameters (direct causality).

COPULA: Gaussian MARGIN 1: Bernoulli MARGIN 2: Bernoulli			
Equation (1): Problem recurrence	est.	t-stat	P-value
Intercept	−0.948	−9.568	0.000
Line cancellation	1.083	5.793	0.000
Bus arrival delay	1.034	4.634	0.000
Bus overcrowding	1.852	14.621	0.000
Long walking distance	−0.929	−3.069	0.002
Level of service satisfaction	−3.118	−6.096	0.000
Equation (2): Expressed frustration	est.	t-stat	
Intercept	−1.152	−6.961	0.000
Problem recurrence	1.837	12.165	0.000
Planning inefficiency	0.491	2.814	0.005
Request for new services	−0.650	−1.759	0.079
Time or monetary loss	1.099	6.788	0.000
Positive attitude (trust that the problem can be resolved)	−0.246	−1.718	0.086
Theta (estimated dependence parameter linking the two equations)	−0.462 [−0.669, −0.216]		
Number of observations	718		
Degrees of freedom	13		
Estimated log-likelihood	−636.5929		
Akaike information criterion	1299.186		
Schwarz (Bayesian) information criterion	1358.68		

Requests for new services are less likely to induce frustration. Passengers who trust that the problem will be resolved are less likely to express frustration. The model shows that both problem recurrence and frustration are not random phenomena and can be explained by the complaint characteristics; hence, understanding transit passengers' frustration helps in prioritizing severe and recurrent problems. The most acute problems are overcrowding, cancellations, and delays, mainly when they generate time or monetary losses. The model shows that maintaining general satisfaction with the level of service and promoting user trust in the system are essential factors in system resilience, since they promote passenger tolerance of service disruptions.

5.2. Number of complaints—spatial distribution

Fig. 6 shows the overall number of complaints relating to unreliability (i.e., line delays and cancellations) for all the lines passing through a particular stop. The map clearly shows that the complaints are not evenly distributed, and some routes and stops are more susceptible to unreliability complaints than others. Such maps help to prioritize stops and lines, as well as to detect problems relating to traffic congestion and traffic light operation (i.e., phase planning and bus preemption) near specific bus stops.

The number of complaints about unreliability (delays and line cancellations) per bus stop, and their spatial distribution as a function of geographic characteristics, were analyzed by estimating a negative binomial model for count data. The explanatory variables were (1) the ratio between the number of passengers boarding daily and the number of daily bus arrivals per bus stop; (2) the traffic congestion level according to the Highway Capacity Manual (2000) level-of-service (LOS) definitions, differentiating between good levels of service (LOS A–C) and low levels of service (LOS D–F); and (3) the distance (in kilometers) on the road network from the central terminal, located near the city center. The number of daily arrivals was calculated from the GTFS files. The levels of service and the motorization levels were obtained from the traffic assignment results of the Jerusalem Transport Model. Table 3 presents the results of the negative binomial model estimation. Because the data showed overdispersion, the negative binomial was superior to the Poisson model. The auxiliary ordinary least squares regression following the Poisson regression yielded $\alpha = 0.825$ with t-statistics equal to 11.52; hence, we can reject the null assumption $\alpha = 0$ and the model is a Poisson model. The likelihood ratio test also shows that the negative binomial model is superior to the Poisson model (LR = 1097.8). The

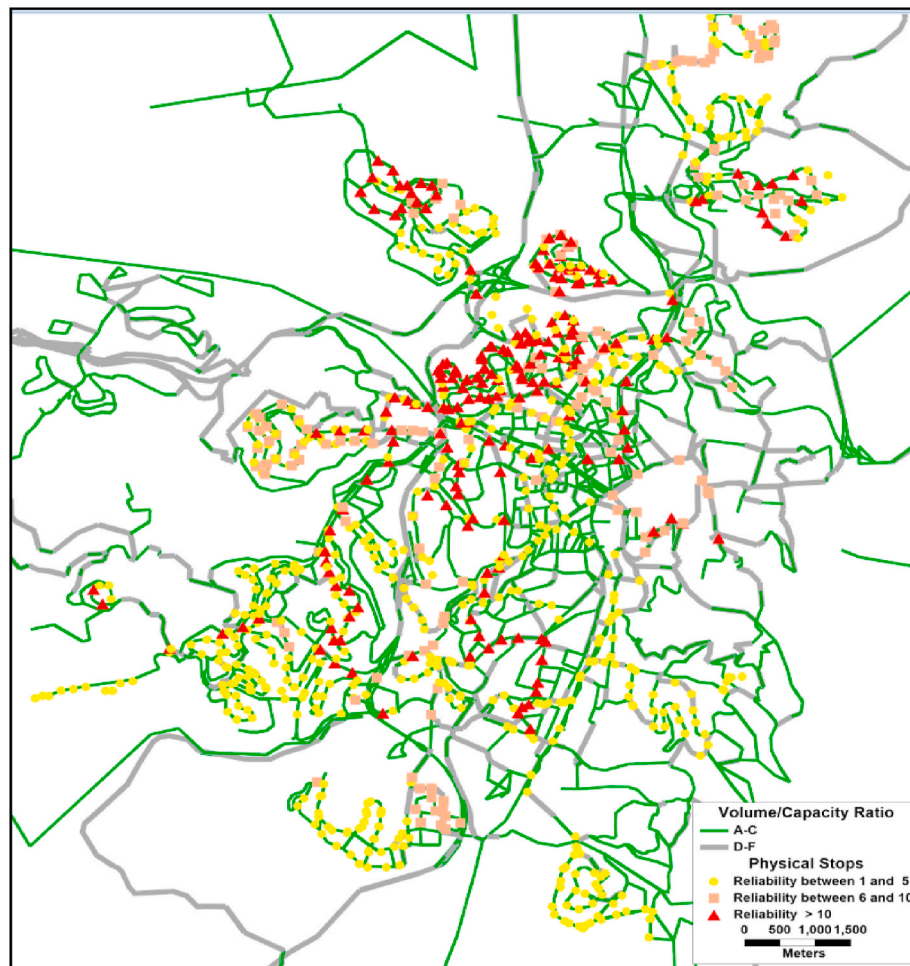


Fig. 6. The spatial distribution of unreliability complaints and traffic level-of-service.

Table 3

Estimated negative binomial model parameters.

Model: negative binomial			
Method: iteratively reweighted least squares (IRLS)			
Covariance: robust			
Dependent variable: Number of unreliability complaints per bus stop	est.	t-stat	P-value
Intercept	2.267	19.63	0.000
The ratio between boarding numbers and number of arrivals	0.005	2.08	0.038
Traffic congestion level: binary indicator. Good level of service (A–C) = 0, low level of service (D–F) = 1.	0.247	1.58	0.115
Distance from the central terminal at the city center (kilometers)	−0.118	−6.69	0.000
Motorization level	−0.001	−2.68	0.007
Number of observations for the model estimation	529		
Degrees of freedom residuals	524		
Degrees of freedom model	4		
Estimated log-likelihood	−1362.2		
Constrained model (Poisson)	−1911.1		
Deviance	613.95		
Pearson Chi ²	509.0		

model results match the spatial spread in Fig. 6. The general traffic congestion level is significant only at the 12% level. Possibly, since congestion-related delays propagate through the system to non-congested areas, the relationship between the number of complaints and the level-of-service is not significant. The other variables are

significant at the 5% level. The number of complaints increases with the ratio between boarding and bus arrivals, and decreases with the distance from the city center. Possibly, since high transit congestion levels characterizes the city center, and most lines begin or end at the central terminal, this area is more susceptible to transit passenger complaints. It is also possible that, since complaints in the MOT's system are recorded per line, it is more difficult to pinpoint the transit delays that derive from traffic congestion and traffic signal operations.

5.3. Complaint classification for bulk handling

The latent class analysis (LCA) yielded five classes according to the Schwarz (Bayesian) information criterion (Appendix Figure A1). The entropy criterion for this solution was 0.97, which indicates reasonably high certainty in the classification (see, e.g., Depaire et al., 2008). We used the average posterior probabilities to calculate the odds of correct classification for each class and measure the accuracy of the classification (for details, see Lanza et al., 2007). The confusion matrix illustrates that over 95% of the observations were correctly identified (Appendix Table A1).

Table 4 presents the distribution of the observed characteristics across the five latent classes. In terms of prevalence, classes 1 and 4 are the largest classes, with 37.1% and 35.2% of the observations, respectively. In terms of both problem recurrence and frustration, class 1 is the most acute, capturing 72.9% of the recurrent problems and 55.8% of the frustration. Class 3, with 11.6% of the observations, captures 17.3% of the recurrent problems and 19.0% of the frustration. Reported tangible loss is divided between class 1, class 3, and class 4, with each capturing

Table 4

Latent class characteristics. Each row sums to 100%.

Variable (%)	Class 1	Class 2	Class 3	Class 4	Class 5
Latent class proportion	37.1	4.0	11.6	35.2	12.1
Line cancellations	0.0	3.2	75.8	12.6	8.4
Delays	30.1	5.5	54.8	8.2	1.4
Overcrowding	66.8	6.4	7.8	12.6	6.4
Low frequency	44.5	3.8	9.9	32.9	8.9
Cleanliness	56.3	12.5	0.0	12.5	18.8
Planning inefficiency	25.0	5.6	1.4	59.7	8.3
Bus stop walking distances	4.2	1.4	0.0	80.3	14.1
Requests for new services	2.3	0.0	0.0	65.9	31.8
Tangible losses	33.7	1.9	30.8	27.9	5.8
Good levels of service perception	2.4	0.0	4.9	0.0	92.7
Positive attitudes indicating trust	41.2	0.0	9.3	39.0	10.6
Experiencing recurrent problems	72.9	7.8	19.0	0.3	0.0
Expressed frustration	55.8	9.0	17.3	14.7	3.2

one-third of the total number of observations.

Class 1 captures multiple problems, with the most prominent being delays, overcrowding, low frequency, and cleanliness. Class 2 includes a small fraction of the complaints, and an overrepresentation of cleanliness complaints. Class 3 captures mainly line cancellation and delay complaints. Class 4 captures most of the complaints about low frequency, planning inefficiency, long walking distances, and new service requests. Class 5 mostly concerns new service requests. In terms of trust and perceptions of high levels of service, class 5 captures all the complaints, including comments about high perceived levels of service. Class 1 and Class 3 together capture 80% of the complaints (divided equally between the two classes), including positive attitudes that indicate trust. The LCA allows the prioritization of complaints by prevalence, severity, planning issues, and the willingness to promote passenger trust. Cluster 1 is the most acute cluster to be resolved based on prevalence and severity. Cluster 4 should be prioritized based on prevalence and promoting passengers' trust. Interestingly, cluster 3, focusing on delays and cancellations, is second in priority to cluster 1, whereas it is currently prioritized as first by the MOT.

Fig. 7 displays the spatial distribution of the classes for lines, addressed in five or more complaints from the same cluster. Cluster 2 contains very few complaints and thus does not appear in the maps. The classes are distributed across several lines and exhibit a diverse spatial structure. As the maps show, clusters 1 and 4 are spatially distributed, whereas clusters 3 and 5 are limited in their spatial spread; hence, resolving clusters 1 and 3 requires system-wide solutions, whereas resolving cluster 3 requires solutions mainly for the central city area. By contrast, cluster 5 calls for solutions for lines leading from the eastern and southern suburban neighborhoods to the city center.

5.4. Limitations

The study provides new insights regarding the use of data science for system-wide analysis and prioritization of bulk complaints. It also identifies the limitations of current complaint-handling practices shared by transit authorities and providers worldwide. First, the complaint data is collected by multiple sources, including the MOT and transit providers, each investigating and answering complaints separately. The number of complaints handled by the MOT represents only a proportion of the total complaints. The bid contracts between the transit operators and the MOT require providers to handle complaints and submit yearly summary reports to the MOT. A unified complaint-handling system is necessary to handle the complaints on a system-wide basis. Second, the current practice of completing an open text box with unstructured and lengthy complaints largely impedes the use of statistical methods for system analysis, due to the need to recode the complaints for entry into a

GIS database. Computerized content analysis can be used to classify and recode the complaints; however, due to the deficiencies of the descriptions, such as incomplete information, word variations, and ironic opinions, such content-based methods are of limited usefulness. A complaint form with location identifiers and closed fields would be more efficient than open text boxes, since it could be designed for compatibility with GIS databases; complaints would then be automatically entered into the GIS database. The current four-stage form is lengthy and cumbersome to use. It elicits passenger contact and personal information at the beginning of the form, which could inhibit complaint submission due to privacy concerns. Eliciting the line and stop information is also relatively complex. The form design prevents the filing of complaints in real time and on site, and location-based information is only partially and retrospectively collected. The result is that complaints can only be analyzed at the line level, disregarding directionality, stop locations, and event locations; thus the current format limits passenger complaints as a big data source. The form does not differentiate between service requests, suggestions, and complaints, and does not ask to rate the overall level of satisfaction. Our study shows that the overall level of satisfaction, positive attitudes, and trust are essential mitigators of transit users' frustration. Furthermore, the current complaint system allows for the reporting of only one incident at a time and does not consider recurrent problems. According to the results of the analysis, information about service problem recurrence should be automatically embedded into the system design.

6. Summary and conclusions

Our study extracts new insights from a system-wide analysis using data science, including mapping, clustering, and statistical models. Implementing the suggested solutions could improve the ability of transport authorities and service providers to receive feedback from passengers and use advanced data analytics to infer system-wide insights. Doing so, transport authorities could prioritize emerging problems, provide system-wide or pinpoint-targeted solutions, and improve both field results and complaint response procedures.

Our results provide practical insights for transit authorities, operators, and planners. Mapping could help planners and authorities to understand the geographical distribution of service disruptions, changes over time, and the impact of service improvements. Mapping complaints by service disruption type or event type can help to identify recurrent issues by line type, operator, employed tangibles, or city regions and provide an estimate of the problem extent. Further research and planning efforts should consider the possibility of combining complaints and traffic data to identify service problems resulting from traffic congestion and signal planning.

Statistical modeling of complaint data can assist in understanding how the characteristics of critical events influence passenger dissatisfaction. In this study, overcrowding and service unreliability were the main concerns, emphasizing recurrent unresolved issues. Because passengers' unpleasant experiences can lead, not only to complaints, but also to temporary ridership discontinuity, it is essential to identify, prevent, and mitigate sources of dissatisfaction. System-wide analysis of passenger complaints and providing adequate solutions should thus enhance user satisfaction for a much larger proportion of passengers than the complaining parties.

Using clustering methods can help to identify different types of complaints and their geographical distribution, recurrence, and prevalence, thus helping to prioritize the complaints according to field-based, rather than arbitrary, criteria. Resolving complaint bulks, instead of handling complaints individually, would increase efficiency and timely customer service. Artificial intelligence (AI) solutions could be applied to increase efficiency and simultaneously handle multiple complaints.

Policy implications accompany the aforementioned practical insights. Our study promotes a change in policymaking to system-wide analysis of user experiences and the use of complaints as valuable

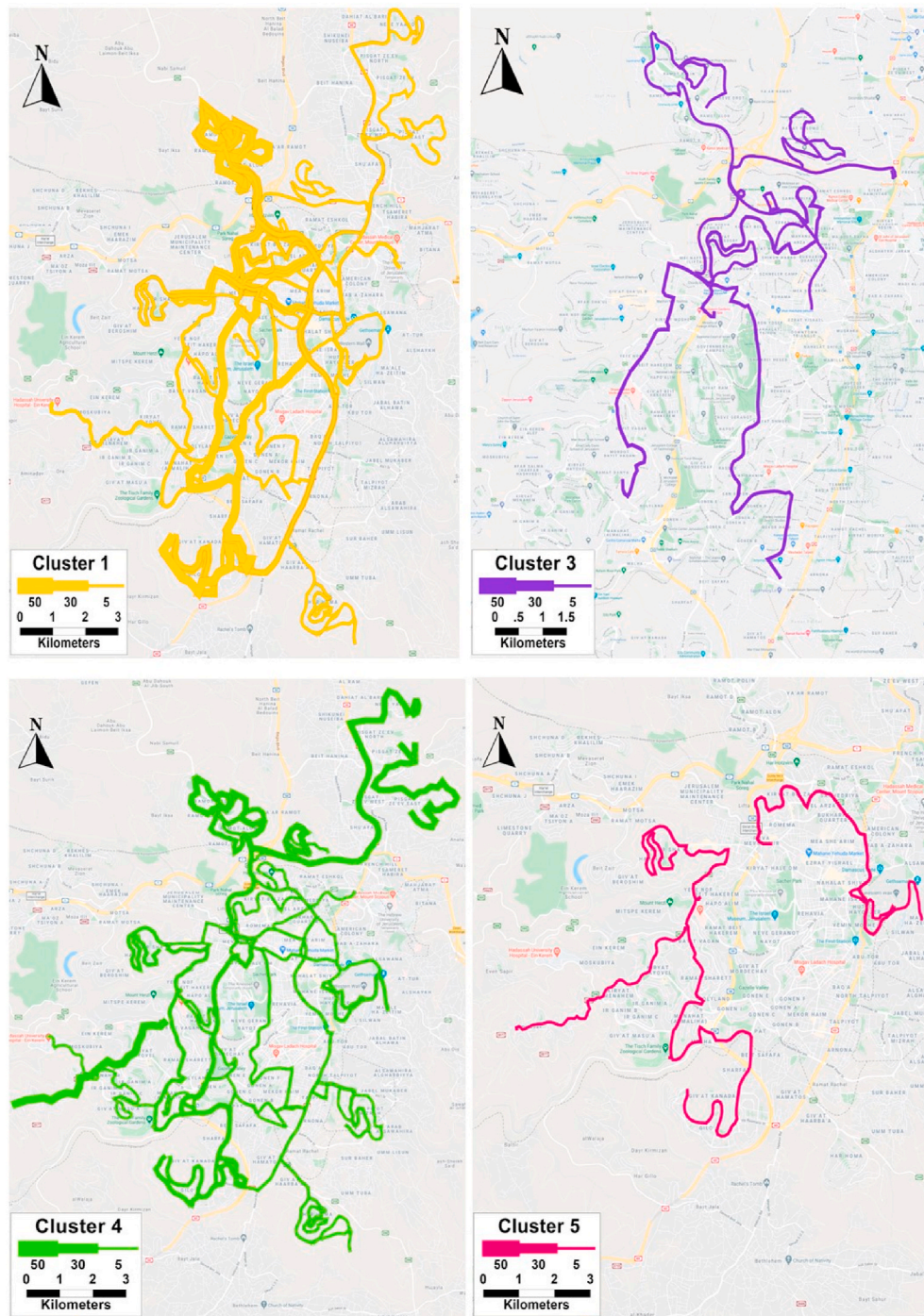


Fig. 7. Spatial distribution of the latent classes.

crowdsourcing data for smart transport systems. First, policymakers should encourage transit passengers to file real-time on-line complaints, which could help to improve transit planning and remedy recurrent issues. Policymakers should also encourage transit providers to promote a favorable complaint culture and facilitate complaint submission. One policy measure that could help in this respect would be to link a complaint culture to the bidding process, for subsidized incumbent operators, franchised services, and private operators. Since better complaint-handling services are related to profit in the private sector (Jonson, 2001), this policy measure also has an economic justification.

Second, complaints should be centralized through a transparent, shared, and visual system, enabling complaints to be handled as

anonymized data points, allowing authorities, passengers, and providers a collaborative tool. Much like maps of traffic congestion and speed, the visualization of complaints should be publicly available. This would allow better route planning by passengers, provide an opportunity to consider spatiotemporal service trends, and aid public participation in discussions about system changes and improvements.

Third, policymakers should take the opportunity to move from ex-post complaint handling to real-time, comprehensive system-monitoring solutions. Policymakers should move from paper-based solutions to high-tech solutions, by setting of policy goals and investments in technology improvements. Current electronic monitoring systems provide technical data, but do not encompass passengers' experiences.

GIS-based complaint systems would provide transport authorities with a system-wide overview of passenger satisfaction and service problems, in addition to other monitoring devices. Such an overview would enable policymakers to better sense the passenger experience. The overview facilitated by technological improvements could lead to real-time passenger-experience monitoring and disruption detection. Real-time online complaint registration and AI could predict and prevent service disruptions. From a policy perspective, passenger complaints should be viewed as a valuable resource, and the extraction of its capabilities can lead to reduce the need for dedicated surveys, improve strategic planning, and to design measures to improve the passenger experience. Last, the study reveals that open-text online complaint-filing practices impede GIS analysis. The main lacunas are the lack of location identifiers and relevant event data, information irrelevance and redundancy, interpretation difficulties, and the inability to automatically feed the data into a GIS database for system-wide analysis. This is not only a technical issue, but has policy implications since, currently, there is a lack of standardized procedures for complaint submission that support system-wide analysis and the use of data-mining and machine learning techniques. A GIS-compatible Smartphone application could facilitate data collection. The user interface could be designed to improve communication with passengers and providers, improve the user experience, and facilitate complaint submission. The proposed tool could increase the efficiency of complaint handling by moving from an individual to a system-wide approach. The proposed tool can also be included as a feature in current transit apps. GIS tools, data analytics, and learning algorithms could be used to detect complaint patterns and predict future service disruptions, remedying the deficiencies of the current complaint submission process by facilitating seamless, location-based, real-time complaint submission and providing GIS-based, system-wide analysis tools. Such a system would save human resources and facilitate complaint handling, prediction, and the prevention of future service disruptions. Such a new system could improve passenger—authority, passenger—operator, and authority—operator communication. (Congosto et al., 2015) (Friman et al., 2001) (Friman, 2004) (Hirschman, 1970) (Jung and Chung, 2016) (Muthén and Muthén, 2007) (Mikhaylov et al., 2016) (Joewono and Kubota, 2007).

Declaration of interest

There is no conflicts of interest.

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